

Muhammad Haseeb

haseeb099m@gmail.com | github.com/ha405 | linkedin.com/in/muhammad-haseeb

RESEARCH INTERESTS

AI Safety; Mechanistic Interpretability; Emergent Misalignment; Domain Generalization; Efficient ML.

EDUCATION

Lahore University of Management Sciences (LUMS)

Sep 2022 – May 2026

B.S. Computer Science

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, AI on Edge Devices, Advanced Topics in ML, LLM Systems, Linear Algebra, Probability.

PUBLICATIONS

- **Mechanistic Evidence for Preserved-but-Misaligned Representations in Non-IID FedAvg (*ICML 2026 Mechanistic Interpretability Workshop*)**.
Muhammad Haseeb, Salaar Masood, Muhammad Abdullah Sohail, Mohammad Fatim Shoaib, Muhammad Tahir.
- **BaCP: Backbone Contrastive Pruning for Preserving Representations in Extremely Sparse Neural Networks (*Preprint*)**.
Mohammad Haroon Khawaja, Muhammad Haseeb, Mohammad Fatim Shoaib, Muhammad Tahir.

RESEARCH EXPERIENCE

ML Research Assistant

Jan 2025 – May 2026

Centre for Urban Informatics, Technology and Policy (CITY), LUMS

Advisors: Dr. Muhammad Tahir, Dr. Zubair Khalid

- **Emergent Misalignment:** Investigating which circuits are associated with misaligned behavior after narrow fine-tuning, whether targeted interventions can reduce it, and whether those interventions transfer across tasks and models.
- **Mechanistic Analysis of Federated Learning:** (*Accepted at ICML 2026 Mechanistic Interpretability Workshop*)
 - Studied FedAvg through the lens of Mechanistic Interpretability to analyze performance loss under Non-IID conditions using class-specific circuit discovery, Universal Sparse Autoencoders (USAEs), and linear probes.
 - Compared class-specific circuits and observed drift and interference patterns during non-IID aggregation, while IID circuits behaved in a perfect constructive manner.
 - Demonstrated via USAE concept analysis that useful latent features, similar to the IID feature basis, remain present and recoverable even when global accuracy degrades under label skew.
- **Backbone Contrastive Pruning (BaCP):**
 - Contributed to a generalized pruning framework to merge standard pruning criteria with contrastive learning to prevent representational collapse.
 - Designed a multi-objective loss function that aligns sparse embeddings with pretrained, fine-tuned, and historical snapshot references to maintain feature consistency with pretrained references to prevent representational collapse.
 - Maintained accuracy comparable to dense baselines at up to **99% sparsity**, outperforming standard unstructured pruning approaches across ViTs/CNNs/Language Models.
- **Domain Generalization & Interpretability:**
 - Applied circuit extraction techniques to isolate task-specific subnetworks, analyzing whether out-of-distribution failures are caused by shortcut learning or weight interference.
 - Optimized visual queries in Vision Transformers using reinforcement learning (Group Relative Query Optimization), achieving a **3% accuracy gain** on the PACS dataset compared to empirical risk minimization.
- **Quantization of Diffusion Models:** Proposed Log-SNR conditioned temporal dynamic quantization (**55% lower LPIPS** on PixArt- α) and implemented Hessian-based GPTQ/QroNos for **4-bit weight compression**.

AI/SWE Research Intern

May 2025 – Aug 2025

University of Illinois Urbana-Champaign (UIUC)

- Investigated LLM-based heuristics for automated `#ifdef` guard insertion in C codebases and explored large language models for software debloating and code dependency analysis.
- Contributed to the development of a VS Code extension supporting a C-to-Rust translation pipeline.

ML Research Assistant

Jan 2025 – Aug 2025

Computer Vision & Graphics Lab, LUMS

- **KL Aware Quantization (KLAQ):** Proposed an augmented GPTQ framework integrating KL divergence and second-order curvature to optimize a combined MSE+KL objective, reducing LLM perplexity by **30%** compared to GPTQ baselines.
- **Single-Image Camera Calibration:** Developed a pipeline to train a Multi-Scale Deformable Transformer by combining loss with geometric constraints to recover full camera projection matrices from a single image.

INDUSTRY EXPERIENCE

Founding Product Engineer

June 2026 – Present

NEXA (Remote)

- Architecting and building highly scalable, distributed backend systems from the ground up.
- Integrating machine learning models and optimizing distributed serving pipelines for fast, efficient inference.

OSS Engineer

Dec 2025 – Feb 2026

DatacurveAI (YC S24)

- Solved bugs and added new features across various open-source machine learning repositories to test and train LLMs on working with large codebases.

Machine Learning Engineer

Jul 2025 – Oct 2025

Innova Tech

- Automated data annotation pipeline, refined training configurations, and fixed model architectures getting a **4% accuracy gain**. Also, implemented inference optimization with TensorRT for speedup and various quantization techniques to reduce memory footprint by **40%** on edge devices.

TEACHING EXPERIENCE

Teaching Fellow, MSAI

Aug 2026 – Present

AI 630: Responsible AI Engineering, LUMS

- Developed and graded quizzes for AI 630: Responsible AI Engineering.
- Managed class logistics and coordinated course projects.

Teaching Assistant

Aug 2025 – Dec 2025

CS436: Computer Vision, LUMS

- Designed and supervised an assignment on transfer learning and multithreaded C++ object detection. Mentored 40 student groups in building a virtual tour application using Structure from Motion.

SKILLS & OPEN-SOURCE

Technical Skills: Python, C, C++, Rust, SQL; PyTorch, Transformers, ONNX, TensorRT, diffusers, adapters.

Open-Source Contributions:

- **pytorch-image-models:** Added F1, precision, and recall metrics to evaluation and training pipelines.
- **adapters:** Implemented PEFT support for Group Query Attention models, fixing tensor mismatch issues.